10 Leaks in underground water pipes are detected using radioactive tracers. Fig. 10.1 shows a radiation detector above a water pipe.

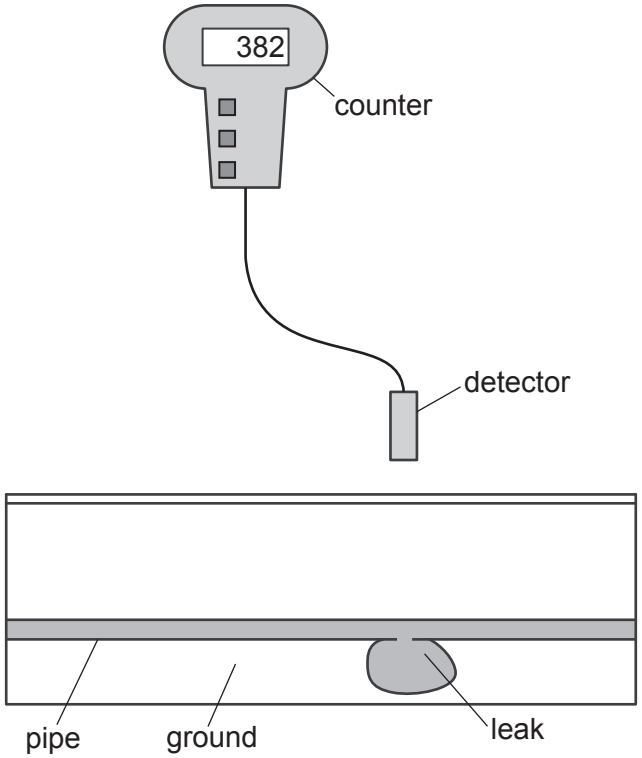


Fig. 10.1

(a) Before the radioactive tracer is added to the water, the detector measures the background radiation above the pipe. The average background radiation is 26 counts / minute.

(i) Define background radiation.

.....

..... [1]

(ii) Suggest **one** source of radiation that may make a significant contribution to the background count rate.

..... [1]

(iii) A radioactive tracer is added to the water. The counter in Fig. 10.1 shows the count rate in counts / minute above the leak in the water pipe.

Determine the count rate due to the tracer.

count rate = ... [2]

© UCLES 2024

(b) Suggest which radioactive emission, alpha, beta or gamma, is suitable for detecting the leak in the water pipe.

Explain your answer.

emission .....

explanation .....

.....

[3]

(c) **(i)** Explain why the radioactive isotope must **not** have a very **short** half-life.

.....

..... [1]

(ii) Explain why the radioactive isotope must **not** have a very **long** half-life.

.....

..... [1]

[Total: 9]